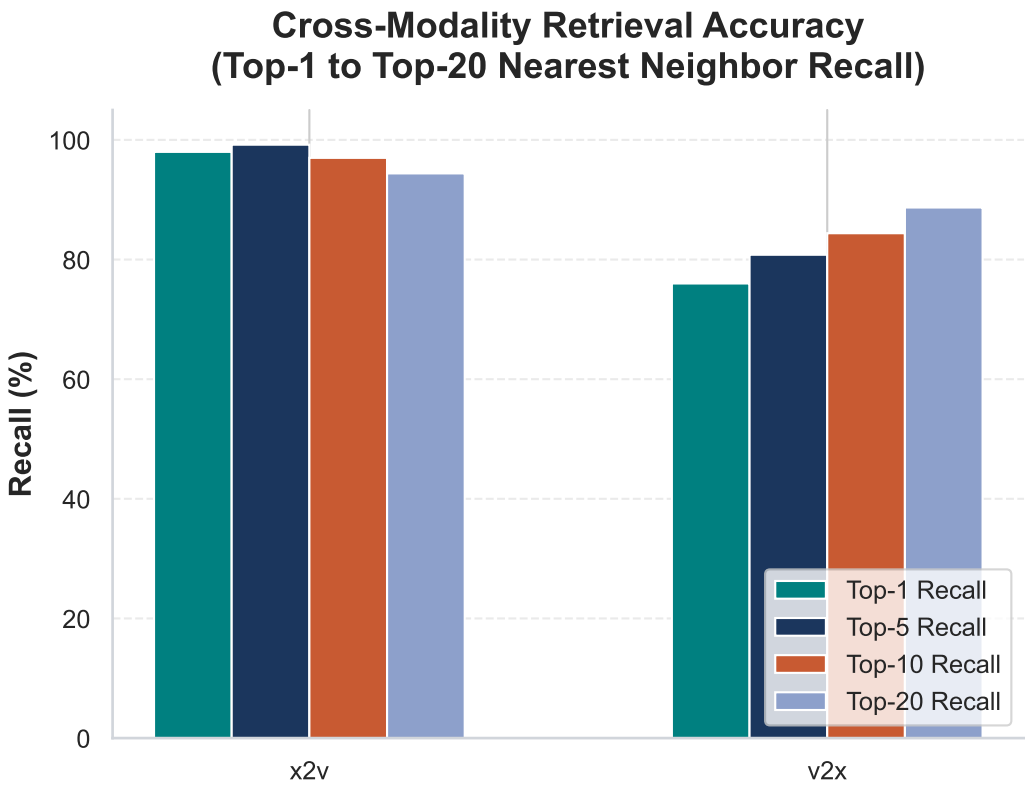
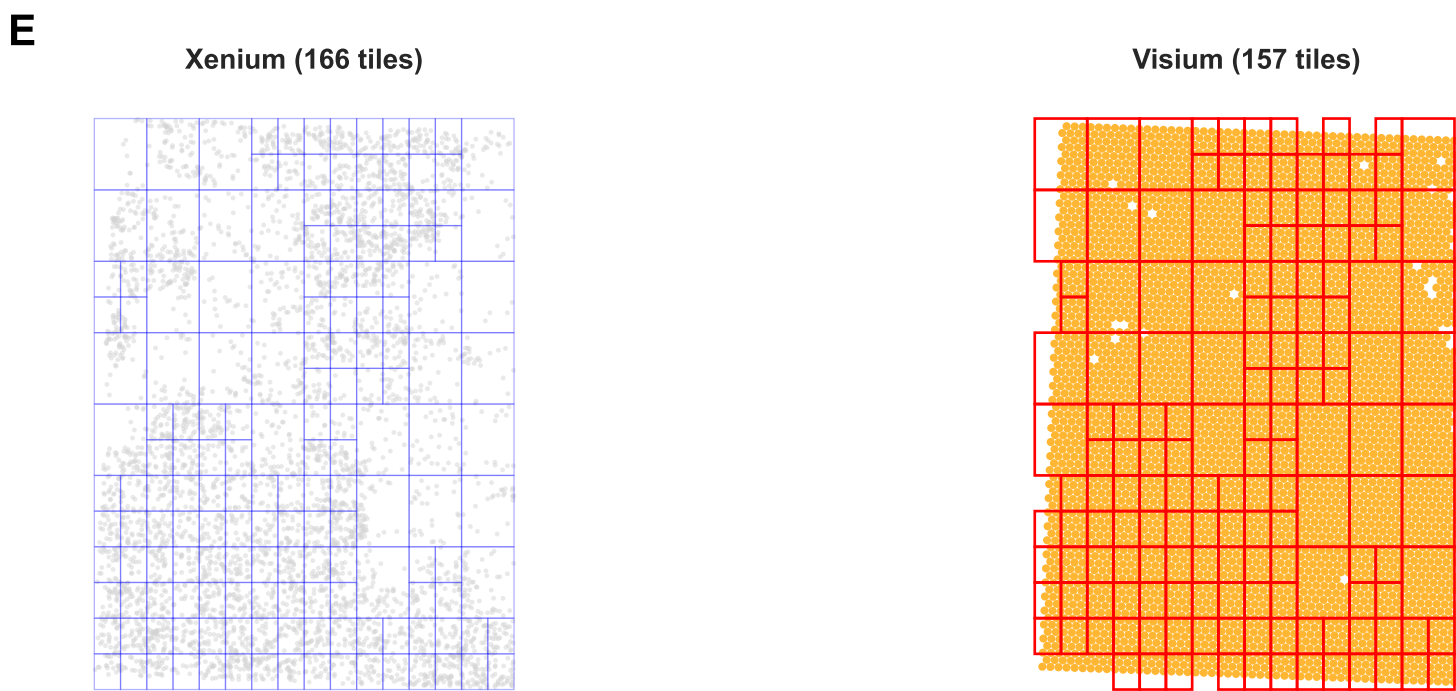
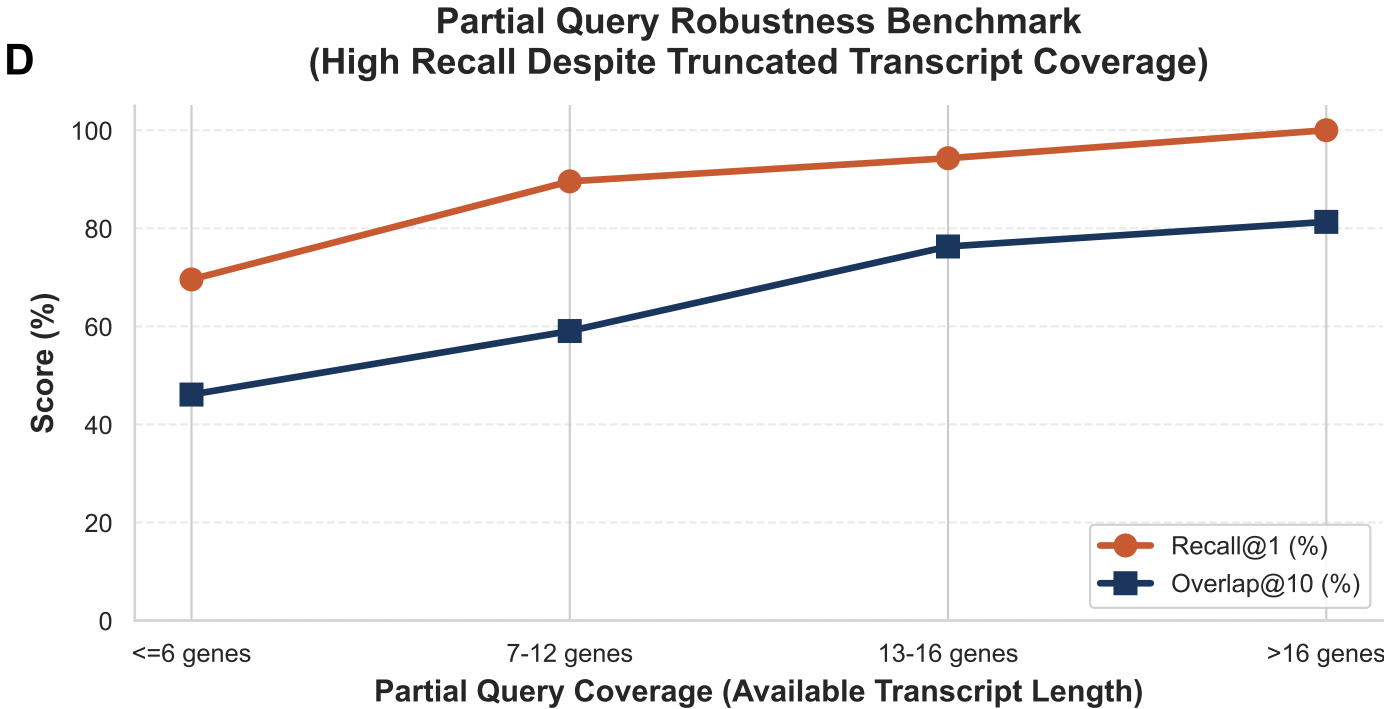
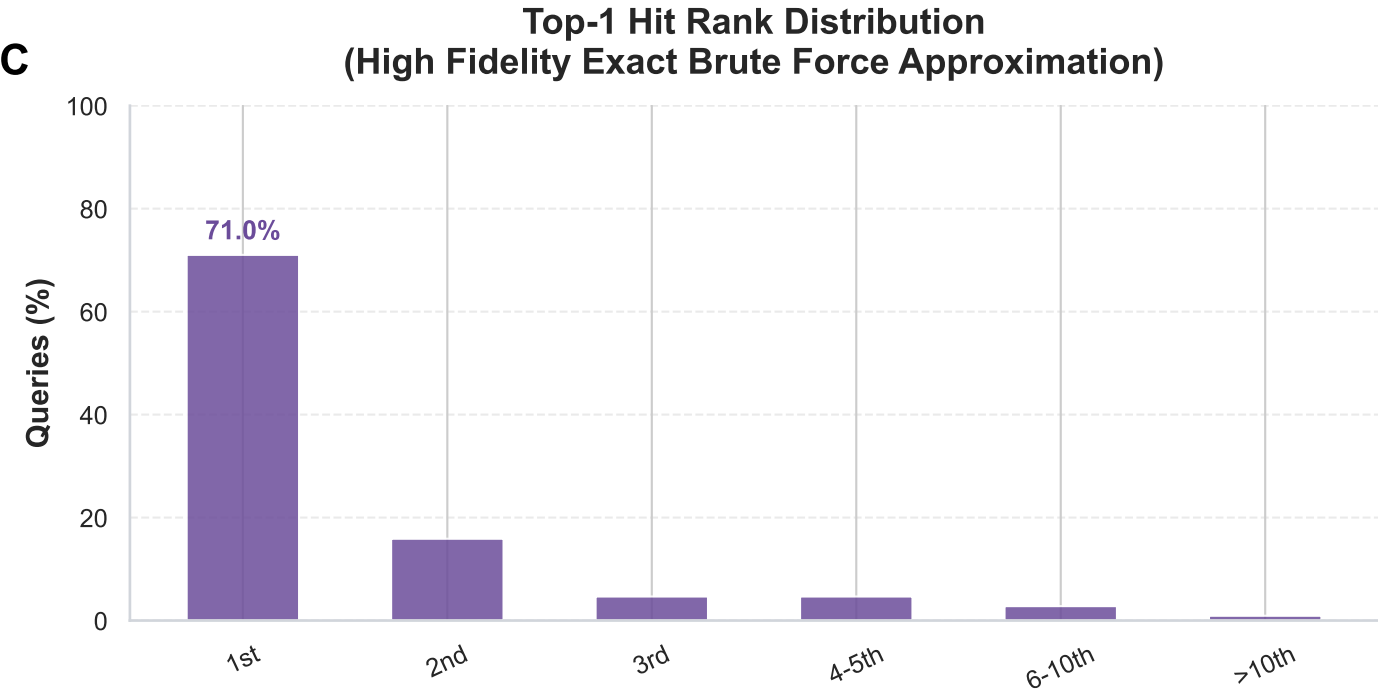
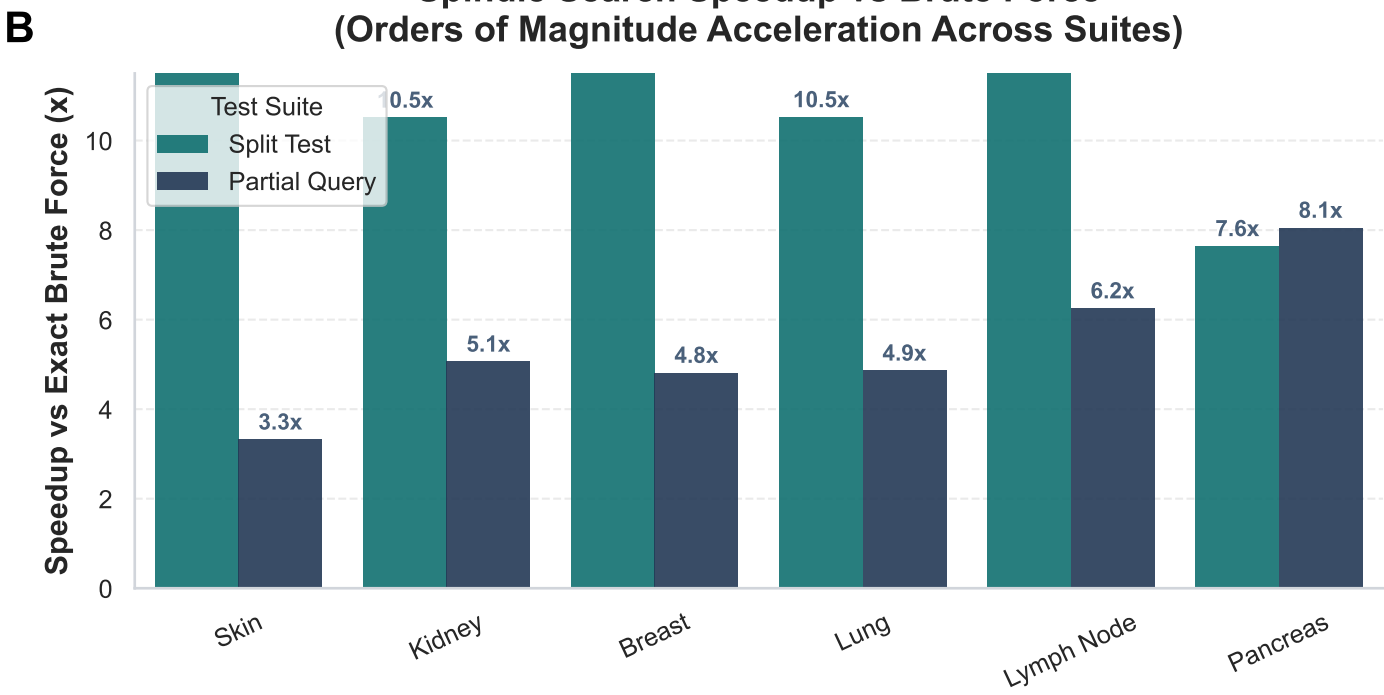
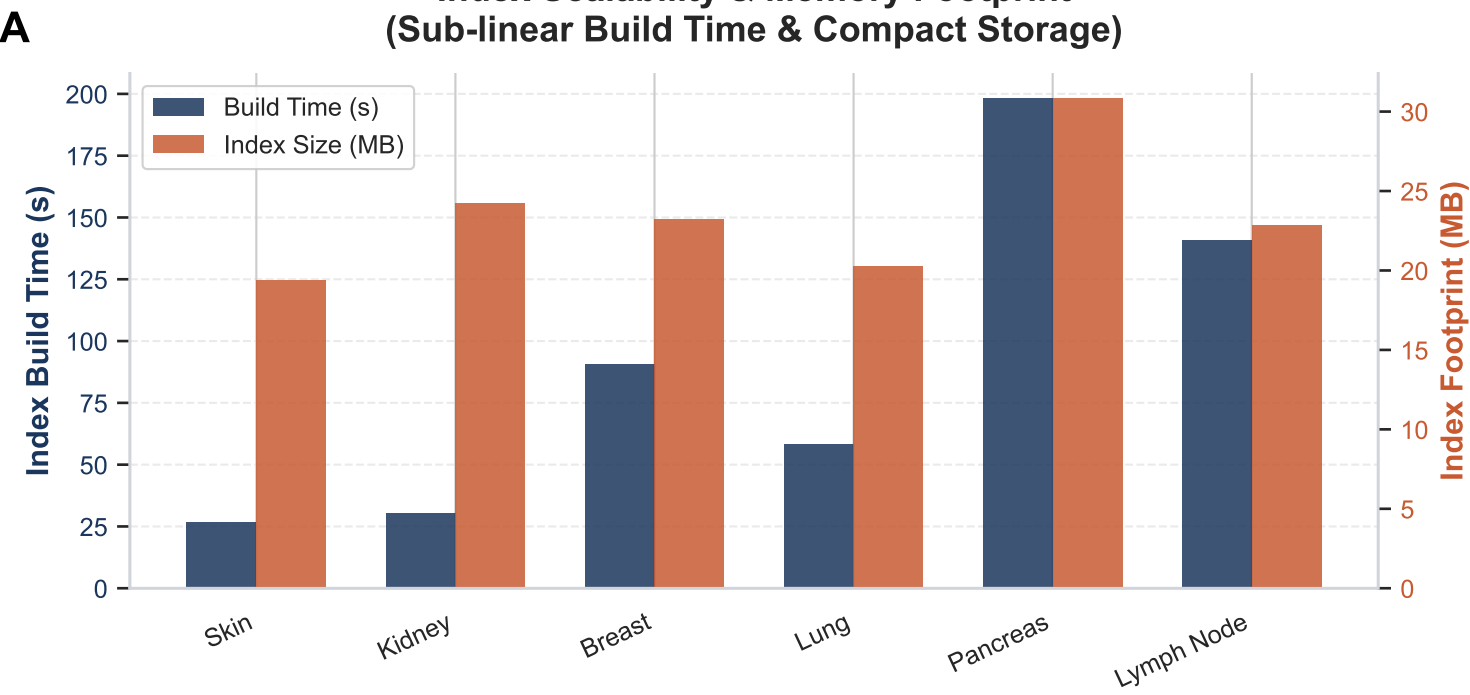
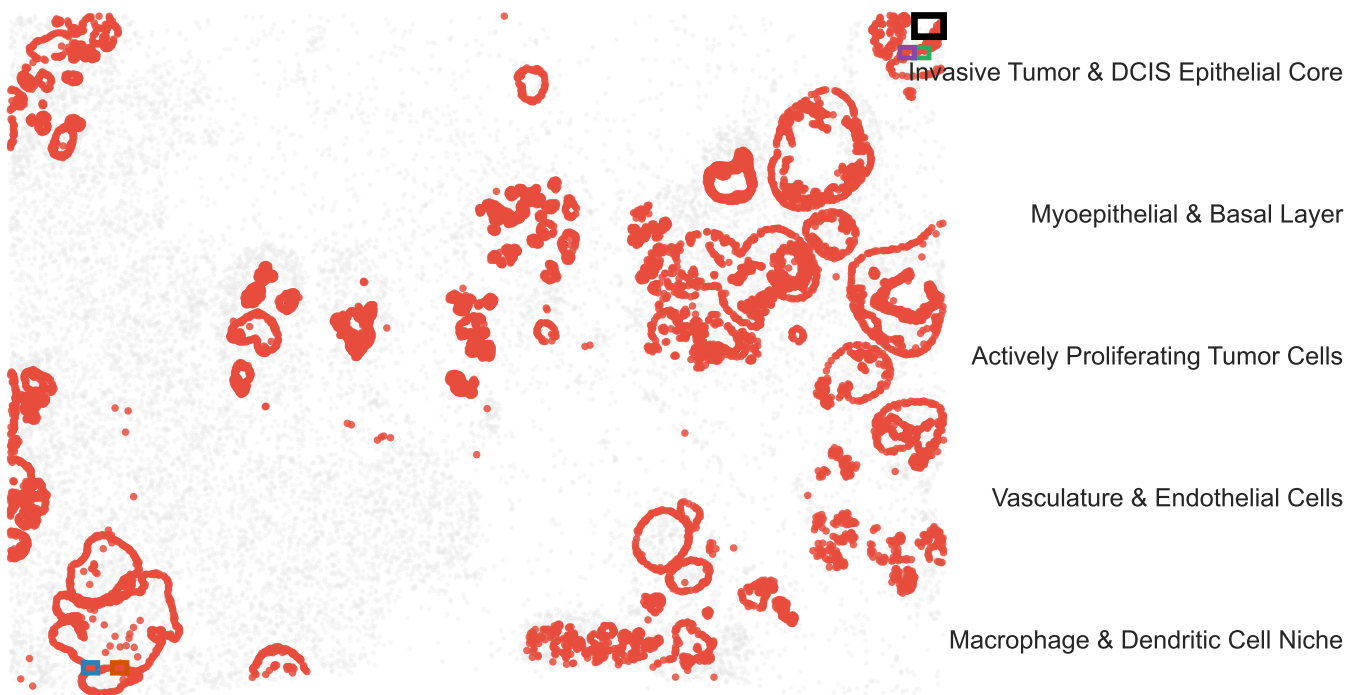


**Figure 1: SPINDLE Core Architecture, Ultra-Fast Indexing, and Spatial Search Performance**  
**Index Scalability & Memory Footprint**  
(Sub-linear Build Time & Compact Storage)



**F**

**Target Niche Spatial Discovery Map**  
(Retrieved Basal Layer Spatial Niche)



**Target Cell Abundance: Background vs. Top 50 vs. Top 10**  
(Spatial Niche Retrieval Enrichment)

